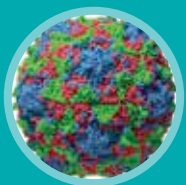


SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

THIS MONTH IN SCIENCE:

This month we cover the wonders of bioluminescence in nature, explore the evolution of sexual reproduction



Researchers capture video of virus as-sembling

Researchers have used a technique known as interferometric scattering microscopy that shows how a virus changes its size over time, and have for the first time seen a virus assembling, even seeing proteins assembling on RNA strands and forming capsids. The study could help scientists fight viruses better, and even create new self-assembling particles. <https://digit.in/virusas-semble>

WHAT'S NEW

Melting permafrost projected to aggravate greenhouse gas emissions

The Arctic regions have been capturing and storing carbon emissions from natural sources for a long time. This CO₂ has been trapped in the permafrost in arctic regions, accounting for 24 percent of the landmass in the northern hemisphere. Studies indicate if greenhouse gas emissions continue to remain unchecked, the melting of the permafrost will accelerate, leading to the release of catastrophic amounts of CO₂. The amount of CO₂ released is expected to exceed what can be used by plants within the year. If greenhouse gas emissions continue at this rate, then the carbon dioxide release from the perma-



frost is expected to go up by 41 percent. Co-author of the study, Dr. Fereidoun Rezanezhad, says "High-latitude cold regions are warming more than twice as fast as the rest of the planet, with the greatest warming occurring during the winter. Given that a major process responsible for CO₂ emissions; microbial respiration, increases with warming, winter is a critical period for carbon cycling".

<https://digit.in/co2pf>

Humpback whale populations on the rise

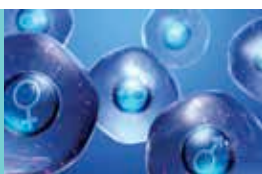
The species of leviathans has almost miraculously come back from the brink of extinction. The whaling industry of the 20th century decimated the humpback whale population in the southern Atlantic, with an estimated 25,000 individuals being caught within a span of 12 years in the early 1900s. The population had at one time diminished to only 450 individuals. According to previous studies, the species had only recovered about 30 percent



of its pre-whaling population. However, newer technologies have shown that the current population is expected to be about 25,000, which is close to the pre-whaling population of humpback whales. The new study

was conducted using ship and aircraft-based surveys as well as computer modelling. The study also looked at how the humpback whale population affects the ecosystem, and vice versa. The whale populations compete with seals and penguins for krill, as their source of food. An increase in temperatures can restrict the krill populations to the south pole, and the range of humpback whales would shrink too.

<https://digit.in/hbwhale>



Male and female mice have different brain cells

Researchers at Caltech investigated brain cells of mice and found that the cells are different in the two sexes, in both cellular composition and gene expression. The finding can be extended to all mammalian brains, including humans. <https://digit.in/braincells>



Magma crystallisation causes explosive volcanoes

Researchers have found that in Plinian eruptions, the rapid ejection of magma causes it to cool and form crystals, which makes the magma more explosive. This was discovered by on site observations and modelling. <https://digit.in/crsmag>



Pushy robots

Researchers at MIT have trained robots as to what to expect when pushing objects. There robots were provided a variety of objects to push, and the reactions to the said objects were then recorded so that the robots could learn the behaviour of pushed objects. The resulting database has been dubbed "omnipush". <https://digit.in/pushybots>